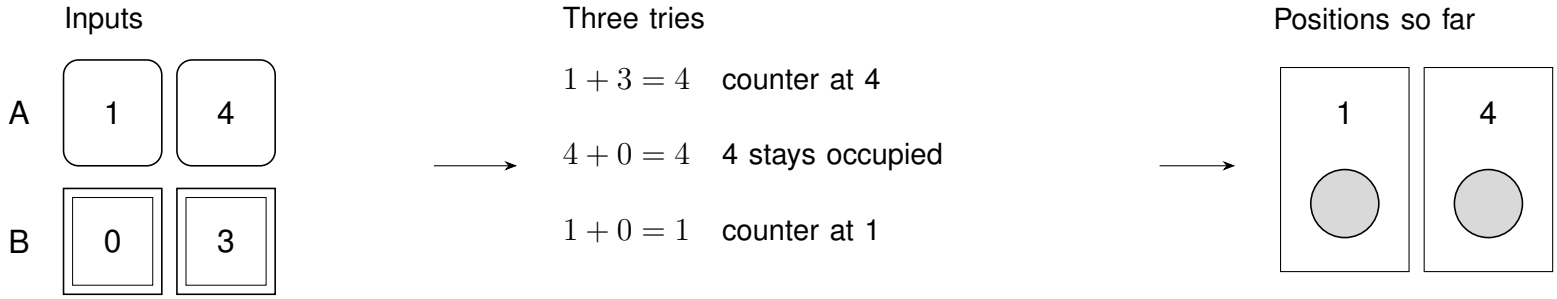
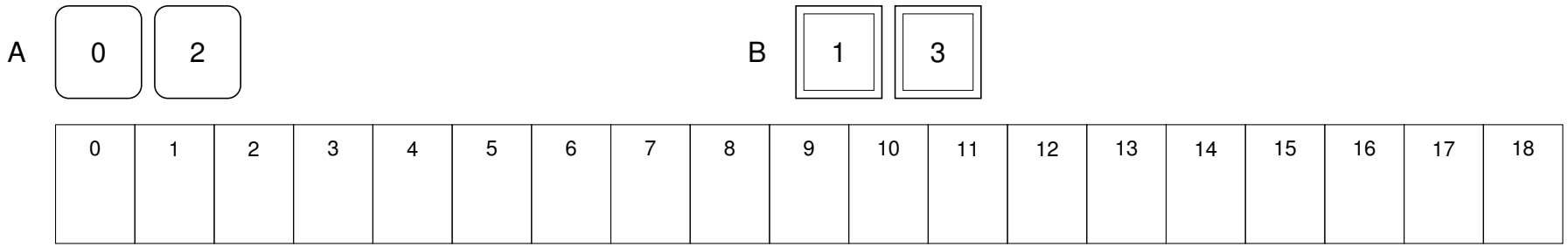
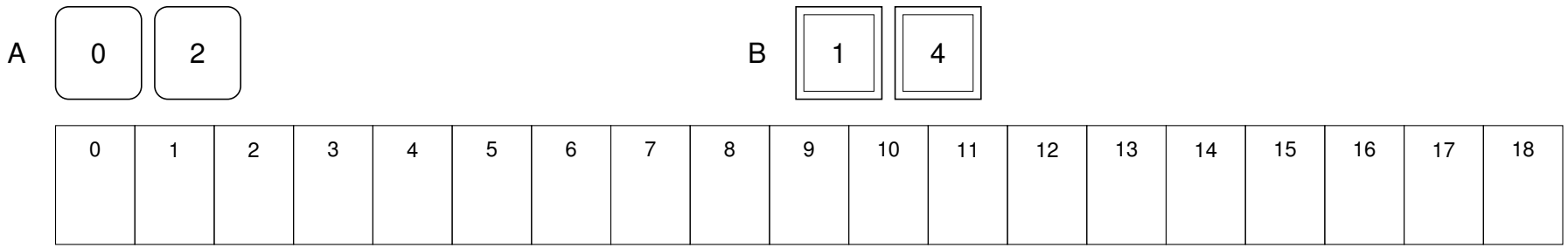
Choose one card from A and one from B. Keep each input fixed while finding all its totals. Put one counter at each different total. Mark occupied cells before reusing counters. No number repeats within an input; A and B may share numbers. Use cards 0–9 unless a problem says otherwise.

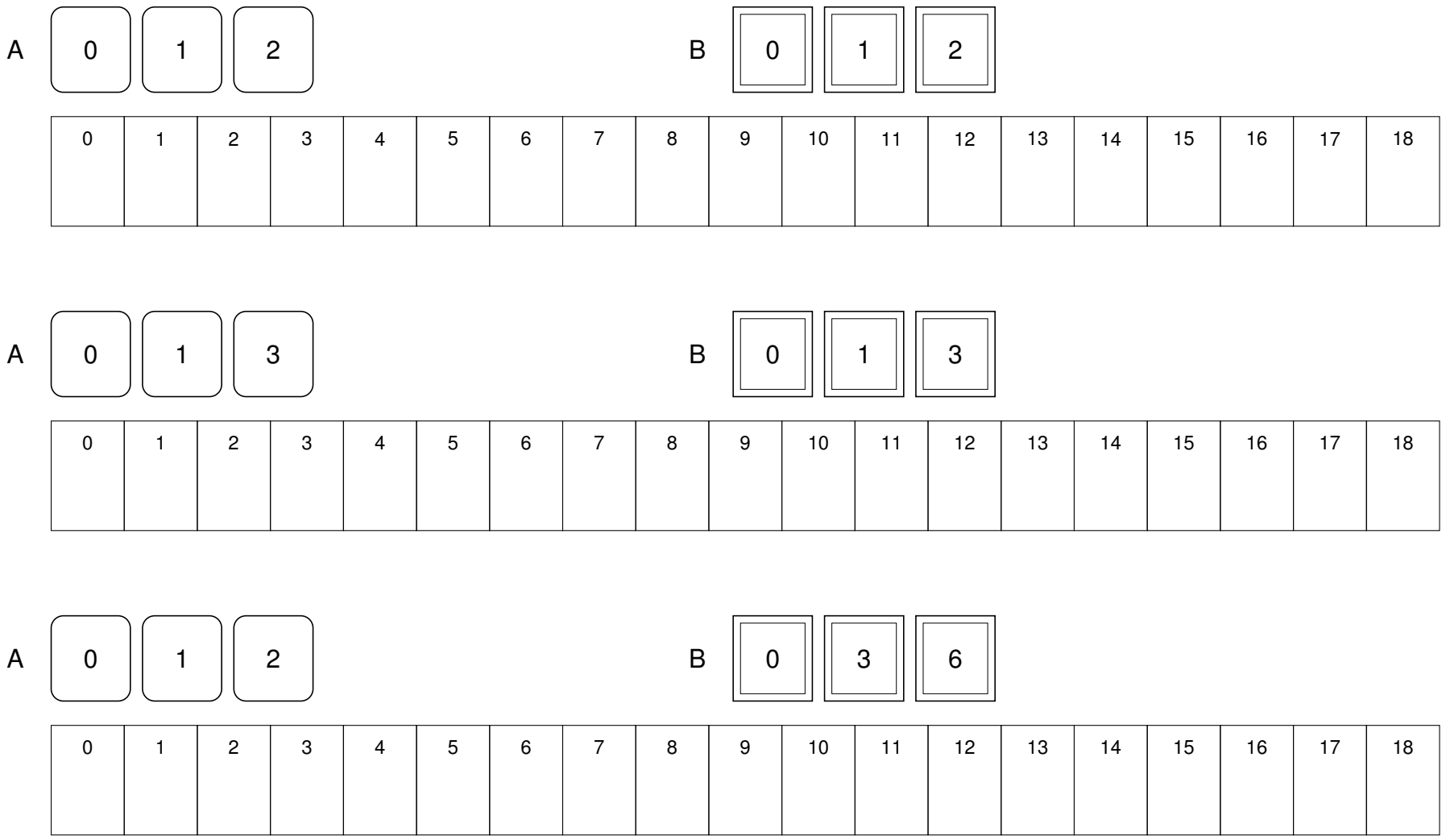



**Problem 1:** Find every different total in each trial. Which trial has fewer?

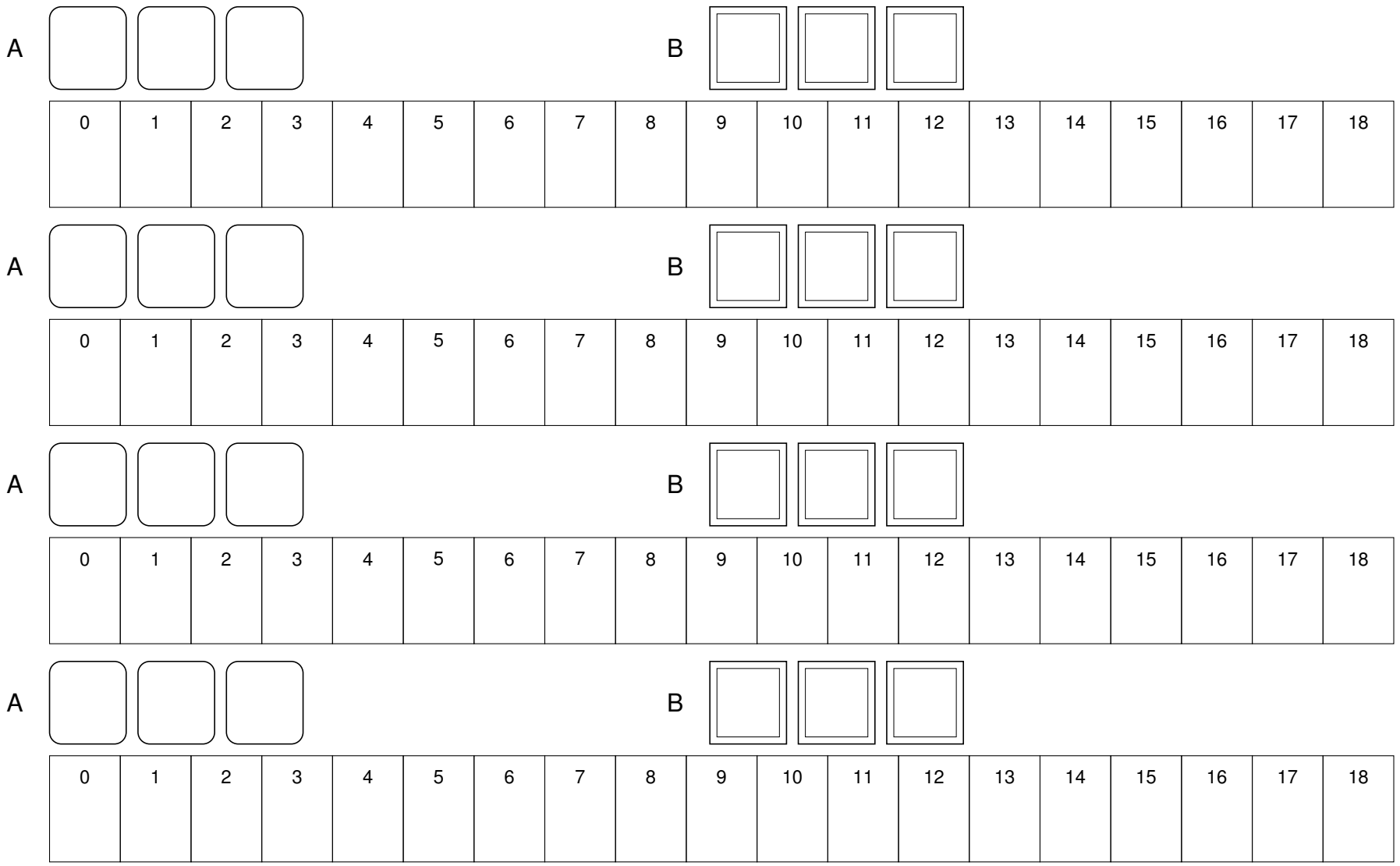


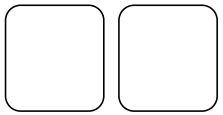
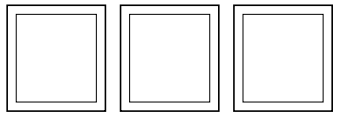
**Problem 2:** Find every different total in these three trials. Which has the fewest? Which has the most?



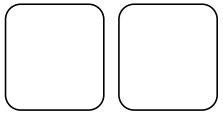
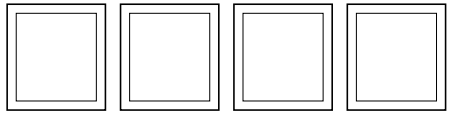
**Problem 3:** Choose three cards for A and three for B. Make as few different totals as you can. Then make as many as you can.



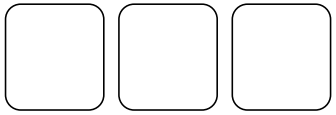
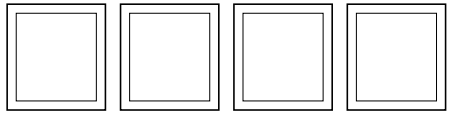
**Problem 4:** For each pair of sizes, choose cards that make the fewest possible different totals. What rule would you predict for other sizes?

A  B 

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | |

A  B 

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | |

A  B 

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | |

**Problem 5:** Keep A fixed. Find every two-card B from 0–9 that gives exactly four different totals.

A | 0 | 3 | 6 |

B | | | Totals: _____

B | | | Totals: _____

B | | | Totals: _____

B | | | Totals: _____

B | | | Totals: _____

B | | | Totals: _____

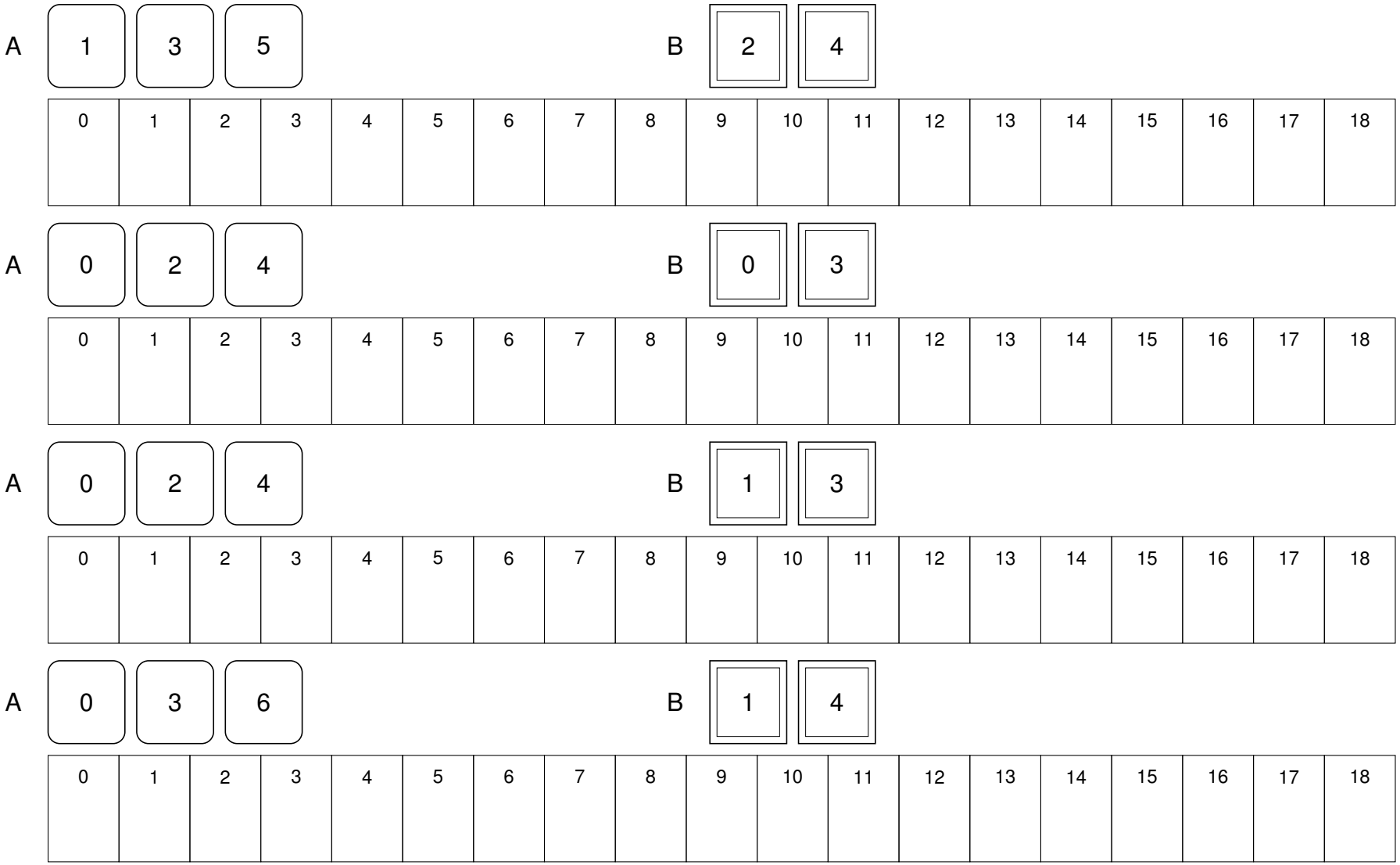
B | | | Totals: _____

B | | | Totals: _____

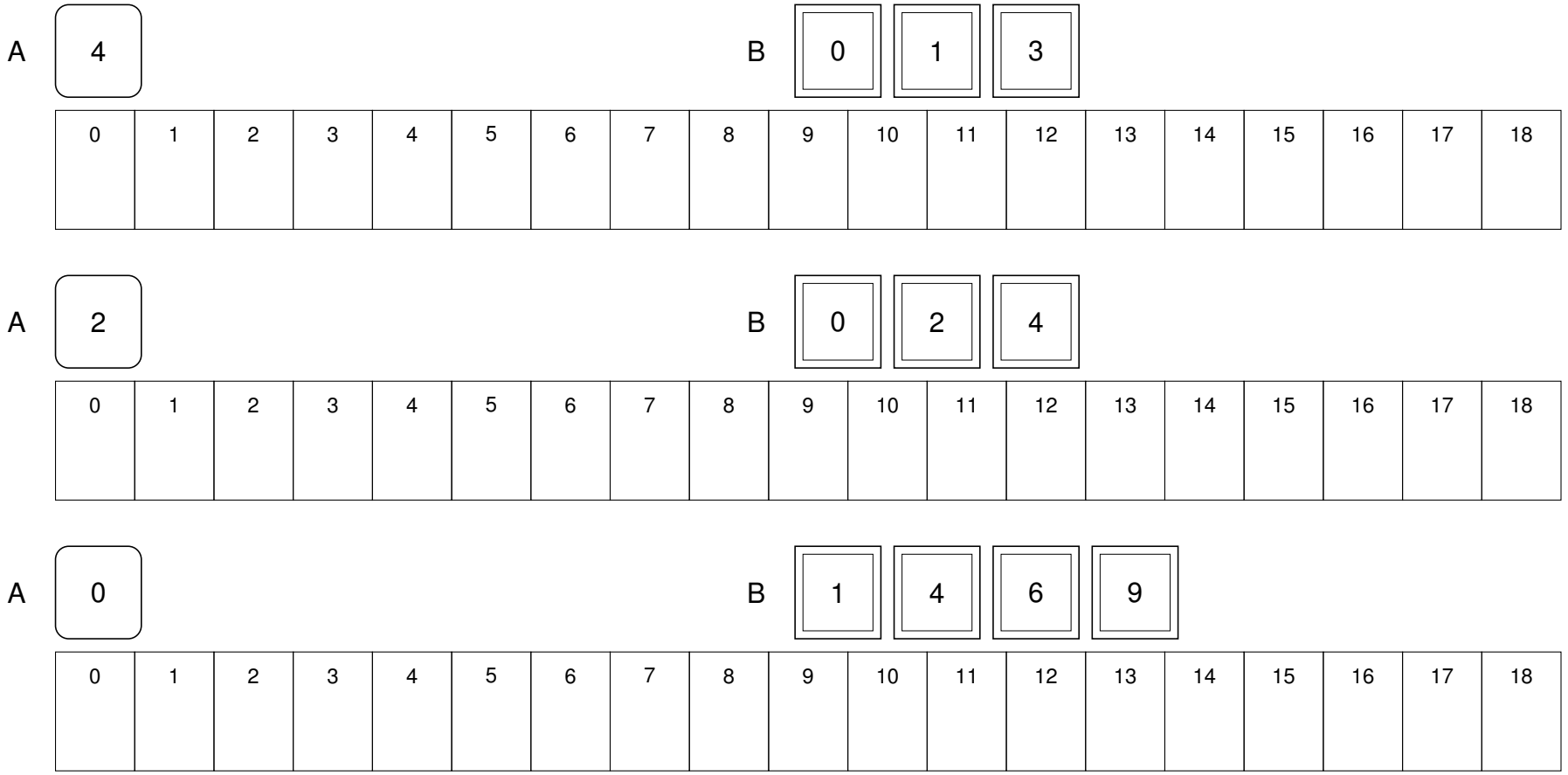
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | |

An input is equally spaced when neighboring numbers in increasing order have the same difference. That difference is its gap.

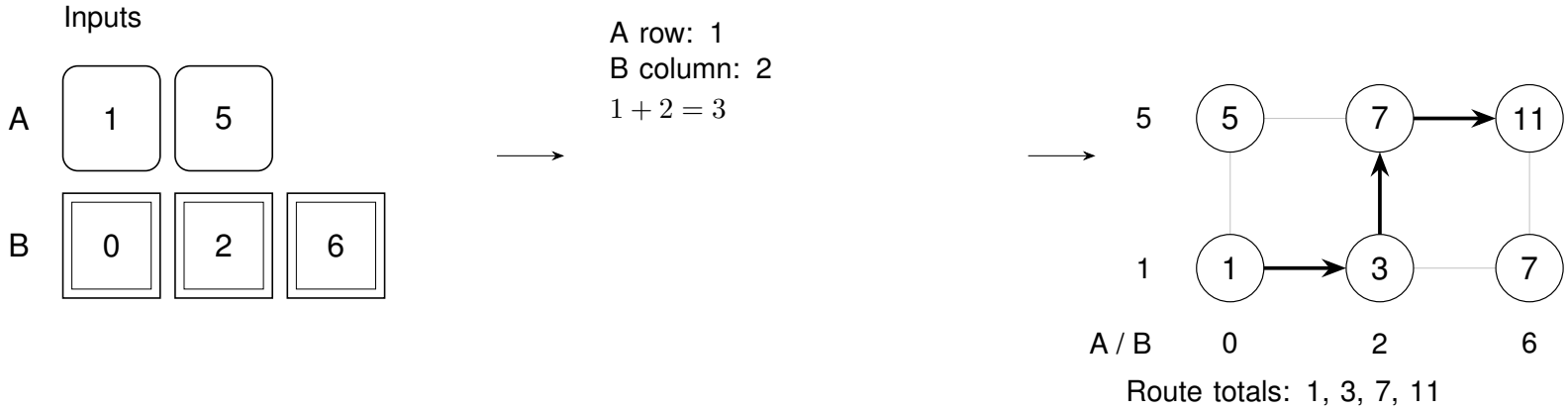
**Problem 6:** Find the different totals for each pair. Each input is equally spaced. Does that always give the fewest possible totals for these sizes?



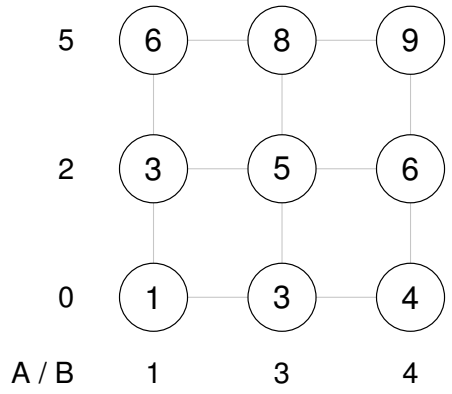
**Problem 7:** Find the different totals. Give a rule for the number of totals when A has one card and B has any number of cards. Does B need equal spacing?

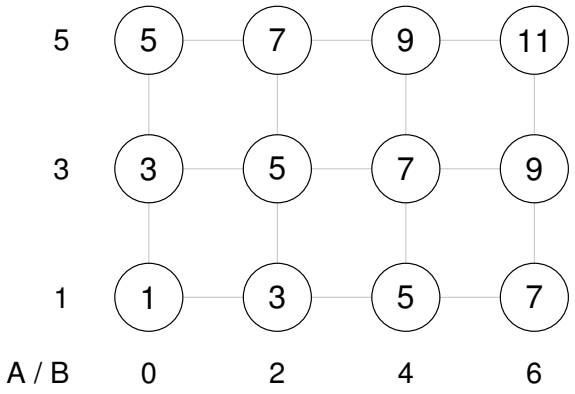


In a grid, A increases from bottom to top and B from left to right. Each dot holds the row card plus the column card. A route starts at the bottom-left dot, ends at the top-right dot, and moves one dot right or one dot up.

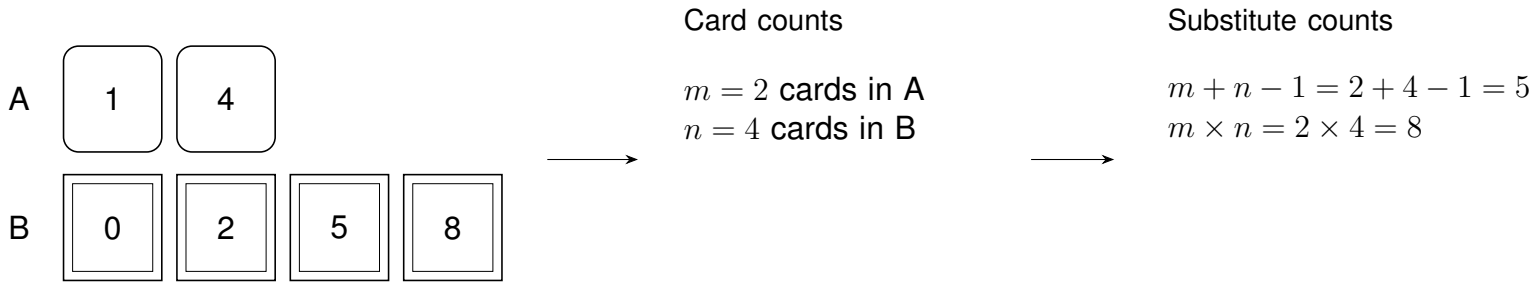



**Problem 8:** Draw several routes in each grid. Can a route repeat a total? How many different totals does every route visit? Explain what makes this happen.



The letters $m$ and $n$ count the cards in A and B.




**Problem 9:** Cards may now be any whole numbers. A has $m$ different cards and B has $n$ different cards; each input has at least one card. Can they have fewer than $m + n - 1$ different totals? Can they have more than $m \times n$? Explain your answers for every possible choice of cards.

**Problem 10:** A has $m$ different whole-number cards and B has $n$. Each input has at least two cards. Explain why two equally spaced inputs with the same gap give exactly $m+n-1$ different totals. Now suppose an A and a B give exactly $m+n-1$ different totals. Must both be equally spaced, with the same gap? Explain why, or find a pair that breaks the claim.